

Laporan Hasil Testing

Report generated on 03-May-2026 at 16:02:48 by [pytest-html](#) v4.2.0

Environment

Python	3.10.0
Platform	Windows-10-10.0.26200-SP0
Packages	- pytest: 8.3.5 - pluggy: 1.6.0
Plugins	- anyio: 4.13.0 - asyncio: 0.25.3 - html: 4.2.0 - metadata: 3.1.1

Summary

158 tests took 00:00:15.

(Un)check the boxes to filter the results.

☐ 0 Failed, ☒ 158 Passed, ☐ 0 Skipped, ☐ 0 Expected failures, ☐ 0 Unexpected passes, ☐ 0 Errors, ☐ 0 Reruns ☐ 0 Retried,

[Show all details](#) / [Hide all details](#)

Result 	Test	Deskripsi	Duration	Links
Passed	tests/chatPipeline_test.py::test_process_query_returns_clarification_when_query_is_ambiguous	-	6 ms	
Passed	tests/chatPipeline_test.py::test_process_query_returns_security_message_when_sql_validation_fails	-	4 ms	
Passed	tests/chatPipeline_test.py::test_process_query_success_without_visualization	-	15 ms	
Passed	tests/chatPipeline_test.py::test_process_query_success_with_visualization	-	19 ms	
Passed	tests/chatPipeline_test.py::test_process_query_falls_back_when_analysis_rate_limited	-	5 ms	
Passed	tests/chatPipeline_test.py::test_process_query_propagates_timeout_http_exception	-	4 ms	
Passed	tests/chatPipeline_test.py::test_process_query_returns_fallback_message_when_llm_unavailable	-	3 ms	
Passed	tests/chatStreaming_test.py::test_handle_chat_returns_stream_with_message_words	-	2 ms	
Passed	tests/chatStreaming_test.py::test_handle_clarification_streams_message_and_keeps_metadata_non_stream	-	2 ms	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_admin_can_access_chatbot_module	Role admin boleh akses module chatbot.	594 ms	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_non_admin_cannot_access_chatbot_module[kepala_divisi]	Role non-admin harus ditolak ketika akses module chatbot.	546 ms	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_non_admin_cannot_access_chatbot_module[karyawan]	Role non-admin harus ditolak ketika akses module chatbot.	546 ms	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_chatbot_module_requires_token	Tanpa token, endpoint chatbot harus return 401.	3 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_kepala_divisi_success	POST chatbot kepala_divisi dengan semua field lengkap harus return 201.	195 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_karyawan_success	POST chatbot Karyawan harus return 201.	190 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_without_addon_prompt	addon_prompt bersifat opsional — boleh null.	190 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_same_otoritas_deactivates_previous_active	Create chatbot aktif baru dengan otoritas sama harus nonaktifkan chatbot aktif lama.	194 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_duplicate_nama_returns_409	Nama chatbot yang sudah ada harus return 409 Conflict.	193 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_duplicate_nama_case_insensitive	Duplikat nama tidak peka huruf besar/kecil.	193 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_missing_nama_returns_422	nama_chatbot wajib — jika tidak ada harus return 422.	189 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_missing_otoritas_returns_422	otoritas wajib — jika tidak ada harus return 422.	188 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_invalid_otoritas_returns_422	otoritas hanya boleh 'kepala_divisi' atau 'karyawan'.	189 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_empty_nama_returns_422	nama_chatbot tidak boleh string kosong.	184 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_nama_stripped_whitespace	nama_chatbot dengan spasi di ujung harus di-strip.	193 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_empty	List chatbot saat DB kosong harus return data= [] dan total=0.	187 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_returns_all_active	Semua chatbot aktif harus muncul di list.	190 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_inactive_included	Chatbot dengan is_active=False tetap muncul di list default.	188 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_pagination_limit	limit membatasi jumlah item per halaman.	193 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_pagination_page_2_different_items	Halaman 2 harus return item yang berbeda dari halaman 1.	193 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_invalid_page_returns_422	page < 1 harus return 422.	187 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_kepala_divisi	Filter otoritas=Kepala Divisi hanya return chatbot Kepala Divisi.	238 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_karyawan	Filter otoritas=karyawan hanya return chatbot Karyawan.	190 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_invalid_otoritas_returns_422	Filter dengan nilai otoritas tidak valid harus return 422.	202 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_search_by_nama	Search berdasarkan nama_chatbot (partial match).	206 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_search_by_addon_prompt	Search juga mencakup konten addon_prompt.	204 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_search_no_result	Search yang tidak cocok harus return data kosong.	201 ms	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_by_id_success	GET /{id} harus return detail chatbot yang benar.	191 ms	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_by_id_not_found	GET dengan ID yang tidak ada harus return 404.	186 ms	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_inactive_returns_200	Chatbot nonaktif tetap bisa diakses via GET /{id}.	188 ms	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_invalid_id_returns_422	ID bukan integer harus return 422.	186 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_nama_only	Update hanya nama_chatbot — field lain tidak berubah.	199 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_otoritas_only	Update hanya otoritas — nama tidak berubah.	208 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_addon_prompt	Update addon_prompt dengan nilai baru.	194 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_addon_prompt_to_null	addon_prompt bisa di-set ke null.	189 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_all_fields	Update semua field sekaligus.	189 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_is_active_to_false	is_active bisa di-update melalui PATCH.	192 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_activate_chatbot_deactivates_other_same_otoritas	Saat aktivasi chatbot, chatbot aktif lain pada otoritas sama harus dinonaktifkan.	207 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_otoritas_active_chatbot_deactivates_active_target_otoritas	Pindah otoritas chatbot aktif harus nonaktifkan chatbot aktif lain di otoritas tujuan.	203 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_nama_to_existing_returns_409	Ganti nama ke nama chatbot lain yang sudah ada harus return 409.	191 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_same_nama_not_conflict	Update ke nama yang sama (milik dirinya sendiri) tidak boleh 409.	194 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_not_found_returns_404	Update chatbot yang tidak ada harus return 404.	187 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_invalid_otoritas_returns_422	otoritas tidak valid saat update harus return 422.	189 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_empty_nama_returns_422	nama_chatbot tidak boleh string kosong saat update.	190 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_returns_200	Soft delete harus return 200 dengan pesan sukses.	190 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_sets_is_active_false	Setelah soft delete, is_active di DB harus False.	192 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_still_accessible_via_get	Setelah soft delete, GET /{id} tetap bisa diakses dengan status nonaktif.	202 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_still_in_list	Setelah soft delete, chatbot tetap muncul di GET / dengan status nonaktif.	197 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_not_found_returns_404	Soft delete chatbot yang tidak ada harus return 404.	197 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_twice_returns_200	Soft delete dua kali bersifat idempotent dan tetap return 200.	213 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_returns_200	Hard delete harus return 200 dengan pesan sukses.	199 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_removes_from_db	Setelah hard delete, record benar-benar hilang dari DB.	194 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_not_found_returns_404	Hard delete chatbot yang tidak ada harus return 404.	187 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_nama_can_be_reused	Setelah hard delete, nama yang sama bisa dipakai lagi.	197 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_default_delete_is_soft	Tanpa ?hard=true, default adalah soft delete — record masih ada di DB.	206 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_detects_ambiguity	Test LLM-based detection mendeteksi ambiguitas.	8 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_clear_query	Test LLM menganggap query yang jelas sebagai tidak ambiguous.	11 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_json_in_markdown_fence	Test parser tetap menerima JSON di dalam markdown code fence.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_tie_breaking_rule	Test tie-breaking rule: score 0.55-0.65 treated as NOT ambiguous.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_api_error_fallback	Test fallback behavior ketika LLM API error.	10 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_invalid_json_fallback	Test fallback ketika LLM return invalid JSON.	8 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationQuestionGenerator::test_generate_question_from_llm	Test generate pertanyaan dari LLM.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationQuestionGenerator::test_generate_question_llm_error_fallback	Test fallback ke template ketika LLM error.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationService::test_process_direct_answer_no_ambiguity	Test direct answer ketika query tidak ambiguous.	22 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationService::test_process_clarification_needed	Test clarification diperlukan ketika query ambiguous.	22 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationService::test_max_clarification_limit	Test max clarification limit per session.	2 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_create_session_context	Test create new session context.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_add_clarification_to_history	Test add clarification to session history.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_scope_preference_storage	Test store scope preference dari clarification answer.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_preference_persistence_across_queries	Test scope preference persist across multiple queries.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_session_ttl_cleanup	Test session TTL cleanup mechanism.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_llm_ambiguous_query	Skenario: LLM mendeteksi ambiguitas dan generate pertanyaan.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_no_ambiguity	Skenario: LLM menganggap query sudah jelas.	10 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_tie_breaking	Skenario: Score borderline (0.55-0.65) apply tie-breaking.	16 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_llm_unavailable	Skenario: LLM tidak tersedia, fallback to safe default.	9 ms	
Passed	tests/kpiGroupCRUD_test.py::test_create_kpi_group_master_with_explicit_sheet_info	Create KPI Group master dengan sheet_id dan nama_grup yang diberikan.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_create_kpi_group_auto_fetch_sheet_info	Create KPI Group auto-fetch sheet_id dan nama dari Google Sheets jika tidak diberikan.	4 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/kpiGroupCRUD_test.py::test_create_kpi_group_tracker	Create KPI Group dengan group_type=tracker.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_all	List semua KPI Groups dengan pagination.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_filter	List KPI Groups dengan filter group_type.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_search	List KPI Groups dengan search keyword.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_tahun_filter	List KPI Groups dengan filter tahun.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_pagination	List KPI Groups dengan pagination calculation.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_get_kpi_group_by_id	Get satu KPI Group dengan all related records.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_get_kpi_group_not_found	Get KPI Group dengan ID yang tidak ada.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_get_kpi_group_master_with_records	Get master KPI Group dengan all master records.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_partial	Update beberapa field pada KPI Group.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_sheet_url_triggers_re_ingestion_master	Update sheet_url pada master group triggers KPI Master re-ingestion.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_tahun_triggers_re_ingestion_master	Update tahun pada master group triggers KPI Master re-ingestion.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_sheet_url_triggers_re_ingestion_tracker	Update sheet_url pada tracker group triggers Tracker re-ingestion.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_master_requires_tahun_when_sheet_url_changes	Update sheet_url pada master group tanpa tahun akan re-ingest dengan tahun=None.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_not_found	Update KPI Group dengan ID yang tidak ada.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_delete_kpi_group	Delete KPI Group by ID.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_delete_kpi_group_commits_transaction	Delete operation commits transaction.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_create_list_get_delete_flow	Complete CRUD flow: create → list → get → delete.	6 ms	
Passed	tests/promptTemplate_test.py::test_build_analysis_prompt_supports_datetime_values	-	0 ms	
Passed	tests/scheduler_test.py::TestSchedulerRepository::test_get_config_returns_default_config	get_config selalu mengembalikan config (default jika file belum ada).	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerRepository::test_update_config_saves_to_manager	update_config harus memanggil save_config dengan perubahan yang diberikan.	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_build_trigger_returns_cron_trigger	_build_trigger harus menghasilkan CronTrigger.	1 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_build_trigger_fires_on_correct_day_and_hour	CronTrigger harus fire pada hari dan jam yang benar tiap bulan.	1 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_auto_pause_if_december_pauses_in_december	_auto_pause_if_december harus remove job dan set is_enabled=False di bulan Desember.	4 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_auto_pause_if_december_skips_non_december	_auto_pause_if_december tidak boleh pause di bulan selain Desember.	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_register_job_enabled_adds_cron_job	register_job harus menambah CronTrigger job saat config enabled.	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_register_job_disabled_does_not_add_job	register_job tidak boleh add job jika scheduler dinonaktifkan.	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_register_job_replaces_existing_job	Jika job lama ada, register_job harus remove lalu add yang baru.	3 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_get_next_run_time_returns_none_when_job_missing	get_next_run_time harus None ketika job scheduler belum ada.	1 ms	
Passed	tests/scheduler_test.py::TestSchedulerController::test_get_config_returns_default	get_config harus selalu mengembalikan config (tidak pernah None).	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerController::test_update_config_updates_and_registers_job	update_config harus simpan perubahan,	3 ms	

Result 	Test	Deskripsi	Duration	Links
		register job, dan isi next_run_at.		
Passed	tests/scheduler_test.py::TestSchedulerController::test_trigger_now_runs_ingestion_job	trigger_now harus mengeksekusi run_ingestion_job.	2 ms	
Passed	tests/userManagement_test.py::TestLogin::test_login_sukses	Login valid mengembalikan access token, refresh token, dan data user.	5 ms	
Passed	tests/userManagement_test.py::TestLogin::test_login_password_salah	Login dengan password salah harus ditolak 401.	4 ms	
Passed	tests/userManagement_test.py::TestLogin::test_login_user_tidak_ditemukan	Login dengan identifier yang tidak terdaftar harus return 401.	3 ms	
Passed	tests/userManagement_test.py::TestLogin::test_login_akun_nonaktif	User nonaktif tidak boleh login dan harus mendapat 403.	4 ms	
Passed	tests/userManagement_test.py::TestLogin::test_login_via_email	Login via email (bukan username) tetap valid dan mengembalikan 200.	4 ms	
Passed	tests/userManagement_test.py::TestLogin::test_login_payload_kosong	Payload login kosong harus gagal validasi request (422).	3 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_sukses_rotation	Refresh token valid harus merotasi token dan menghasilkan token baru.	7 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_token_sudah_direvoke	Refresh token yang sudah direvoke harus ditolak 401.	5 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_token_kadaluarsa	Refresh token kadaluarsa harus ditolak 401.	4 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_token_palsu	Refresh token palsu dengan format JWT tidak valid harus ditolak 401.	3 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_kirim_access_token	Access token dikirim ke /refresh → ditolak karena type mismatch.	18 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_user_nonaktif	Refresh token milik user nonaktif harus ditolak 403.	5 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_user_dihapus	Refresh token untuk user yang sudah tidak ada harus ditolak 401.	6 ms	
Passed	tests/userManagement_test.py::TestRefresh::test_refresh_revoke_dipanggil_sekali	Pada refresh sukses, revoke_token harus dipanggil tepat satu kali.	6 ms	
Passed	tests/userManagement_test.py::TestLogout::test_logout_sukses	Logout valid harus revoke refresh token dan mengembalikan 200.	5 ms	
Passed	tests/userManagement_test.py::TestLogout::test_logout_token_palsu	Logout dengan refresh token palsu harus ditolak 401.	3 ms	
Passed	tests/userManagement_test.py::TestLogout::test_logout_token_kadaluarsa	Logout dengan refresh token kadaluarsa harus ditolak 401.	4 ms	
Passed	tests/userManagement_test.py::TestLogout::test_logout_idempotent	Logout dua kali tidak boleh error (revoke_token bersifat idempotent).	6 ms	
Passed	tests/userManagement_test.py::TestCreateUser::test_create_user_sukses	Admin dapat membuat user baru ketika username/email belum terpakai.	194 ms	
Passed	tests/userManagement_test.py::TestCreateUser::test_create_user_duplikat_username	Pembuatan user harus gagal 409 jika username sudah terdaftar.	5 ms	
Passed	tests/userManagement_test.py::TestCreateUser::test_create_user_duplikat_email	Pembuatan user harus gagal 409 jika email sudah terdaftar.	5 ms	
Passed	tests/userManagement_test.py::TestCreateUser::test_create_user_forbidden_sebagai_kepala_divisi	Kepala divisi tidak boleh membuat user baru (POST /users hanya admin).	3 ms	
Passed	tests/userManagement_test.py::TestCreateUser::test_create_user_forbidden_sebagai_karyawan	Karyawan tidak boleh membuat user baru.	3 ms	
Passed	tests/userManagement_test.py::TestCreateUser::test_create_user_password_lemah	Password tanpa huruf kapital → 422.	4 ms	
Passed	tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_sukses	Admin dapat melihat daftar user dengan metadata total yang benar.	6 ms	
Passed	tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_pagination	Parameter page dan limit harus diteruskan ke repository dengan benar.	5 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_with_search_role_status_filter	Parameter search, role, dan status diteruskan ke service/repository.	6 ms	
Passed	tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_forbidden_sebagai_kepala_divisi	Kepala divisi tidak boleh akses list semua user.	3 ms	
Passed	tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_forbidden_sebagai_karyawan	Karyawan tidak boleh akses list semua user.	4 ms	
Passed	tests/userManagement_test.py::TestGetUserById::test_get_user_by_id_sukses	Admin dapat mengambil detail user berdasarkan user_id yang valid.	4 ms	
Passed	tests/userManagement_test.py::TestGetUserById::test_get_user_by_id_not_found	Pengambilan detail user harus return 404 jika user_id tidak ditemukan.	4 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_hash_dan_verify_password	hash_password menghasilkan hash bcrypt yang lolos verify_password untuk nilai benar.	557 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_create_access_token_berisi_field_wajib	Access token harus memuat sub, username, role, type access, dan expiry positif.	1 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_create_refresh_token_berisi_field_minimal	Refresh token hanya memuat field minimal tanpa role/username.	1 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_decode_access_token_tolak_refresh_token	decode_access_token harus menolak token bertipe refresh.	2 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_decode_refresh_token_tolak_access_token	decode_refresh_token harus menolak token bertipe access.	1 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_decode_token_kadaluarsa	decode_access_token harus melempar 401 untuk token yang sudah kadaluarsa.	8 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_authenticate_user_sukses	authenticate_user mengembalikan user saat identifier dan password valid.	14 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_rotate_tokens_revoke_lama_terbit_baru	rotate_tokens harus revoke token lama dan menerbitkan pasangan token baru.	18 ms	
Passed	tests/userManagement_test.py::TestAuthServiceUnit::test_rotate_tokens_reuse_ditolak	rotate_tokens harus menolak refresh token yang terdeteksi reuse/revoked.	65 ms	